

Centra-SD

Centralized Network Performance Monitoring and Device Management Platform

Technical Product Documentation

1. Introduction

Overview

Centra-SD is a centralized network performance monitoring and management platform designed to simplify the supervision, control, and analysis of large-scale wireless network testing operations. It serves as the central server that collects measurement data from multiple field test devices, consolidates the information into a unified repository, and presents the results through an intuitive web-based interface.

Modern mobile networks consist of thousands of base stations, numerous service areas, and a variety of radio access technologies. Monitoring network performance across such large infrastructures requires continuous testing and efficient management of field devices. Centra-SD addresses this challenge by enabling engineers to remotely control test devices, automate measurement schedules, monitor device status in real time, and analyze Key Performance Indicators (KPIs) through interactive dashboards.

Unlike traditional monitoring systems that require manual collection of log files from individual devices, Centra-SD automatically retrieves uploaded measurement data and stores it within a centralized database. This centralized architecture significantly reduces operational complexity while improving visibility across the entire testing network.

The platform is designed to work seamlessly with field measurement applications such as **Sigma-ML**, allowing organizations to perform large-scale automated testing campaigns with minimal manual intervention. Through browser-based access, network engineers can manage testing operations from virtually any location with internet connectivity.

2. Purpose of Centra-SD

The primary objective of Centra-SD is to provide a centralized platform for managing distributed wireless network testing operations. As telecom networks continue to expand, organizations require efficient methods to coordinate multiple test devices, monitor network quality, and analyze collected measurements.

Centra-SD fulfills these requirements by:

- Consolidating measurement data from multiple devices into a single repository.
- Providing centralized control over remote test devices.
- Automating test scheduling and execution.
- Monitoring network performance in real time.
- Supporting historical analysis of collected measurements.

- Simplifying KPI visualization through dashboards, charts, and maps.
- Reducing manual effort associated with field testing.
- Improving operational efficiency through automation.
- Enabling quick identification of network bottlenecks and service degradation.

By integrating automation, remote management, and centralized analytics, Centra-SD enables organizations to maintain consistent testing processes across geographically distributed locations.

3. Product Objectives

Centra-SD has been developed with several strategic objectives that support efficient wireless network management.

Centralized Management

Provide a single platform for controlling multiple field test devices across different geographic regions.

Real-Time Monitoring

Display current network performance and device status as measurements are collected.

Automated Operations

Reduce manual effort by automating test scheduling, data collection, and report generation.

Data Consolidation

Store measurement logs from multiple devices within a centralized repository for easy access and long-term analysis.

Efficient Analysis

Provide interactive visualization tools that simplify KPI evaluation and help engineers identify network issues quickly.

Scalability

Support deployment across organizations managing hundreds or thousands of remote test devices.

4. System Architecture

Centra-SD follows a centralized client-server architecture.

5. Working Principle

Centra-SD operates as the central intelligence layer of a distributed testing infrastructure.

The workflow consists of the following stages:

Step 1 – Device Registration

Each field test device is registered within the Centra-SD platform. Devices can be organized into logical groups based on region, project, operator, or testing purpose.

Examples include:

- North Region
 - South Region
 - LTE Optimization Team
 - 5G Rollout Project
 - Drive Test Vehicles
-

Step 2 – Schedule Creation

Network administrators create test schedules through the web interface.

Typical test schedules include:

- FTP Download
- FTP Upload
- HTTP Download
- Voice Calls
- SMS Testing
- Ping Tests
- Continuous Data Sessions

Schedules define:

- Start Time
 - End Time
 - Test Frequency
 - Device Assignment
 - Measurement Duration
-

Step 3 – Remote Deployment

Using push technology, Centra-SD distributes the configured schedules to selected test devices in real time.

Once received, the devices automatically begin executing the assigned measurements without requiring user interaction.

Step 4 – Data Collection

During execution, each test device uploads measurement logs to the Centra-SD server.

Collected information includes:

- Signal Strength
- Throughput
- Latency
- GPS Coordinates
- Voice Quality
- Protocol Messages
- Device Status

The server stores this information within a centralized repository for immediate visualization and future analysis.

Step 5 – Visualization and Analysis

As measurement data is received, Centra-SD updates dashboards, maps, and KPI charts in real time.

Engineers can monitor:

- Device locations
- Test progress
- Network coverage
- KPI trends
- Service quality
- Problem areas

Interactive filtering and drill-down capabilities allow users to investigate individual measurements or evaluate overall network performance.

6. Core Features

Centra-SD provides a rich set of capabilities that simplify large-scale network testing operations.

6.1 Centralized Device Management

One of the most significant features of Centra-SD is its ability to manage all field test devices from a single web-based platform.

Administrators can:

- Register new devices.

- Remove inactive devices.
- Create device groups.
- Monitor device status.
- Track connectivity.
- View active test sessions.
- Push configurations remotely.

This centralized management eliminates the need to configure devices individually in the field.

6.2 Browser-Based Access

Centra-SD is accessible through any standard web browser, allowing authorized users to monitor testing operations from anywhere with internet access.

Benefits include:

- No client software installation.
 - Platform independence.
 - Remote accessibility.
 - Centralized administration.
 - Simplified deployment.
-

6.3 Interactive User Interface

The platform provides an intuitive graphical interface designed to simplify network monitoring.

Key interface components include:

- Dashboard
- KPI Panels
- Interactive Maps
- Charts and Graphs
- Device Status View
- Schedule Manager
- Historical Reports

These components enable engineers to quickly identify abnormal network behavior and investigate performance issues.

6.4 Automation Engine

Automation is a fundamental capability of Centra-SD.

The built-in scheduling engine allows organizations to automate repetitive testing tasks across multiple devices simultaneously.

Automation supports:

- Daily schedules
- Weekly schedules
- Continuous monitoring
- Event-based execution
- Multiple concurrent tests

This reduces operational effort and ensures consistent measurement procedures across all field devices.

6.5 Central Repository

All uploaded measurement data is stored in a centralized repository, providing a single source of truth for network performance analysis.

Benefits include:

- Simplified data retrieval
- Long-term storage
- Historical trend analysis
- Improved collaboration
- Centralized backup
- Consistent reporting

7. Automation and Scheduling Engine

Automation is one of the most important capabilities of Centra-SD. Managing hundreds of test devices manually is time-consuming and prone to inconsistencies. Centra-SD addresses this challenge by providing a powerful scheduling engine that automates network testing activities across multiple devices simultaneously.

Instead of manually starting tests on each device, administrators can create reusable test schedules and assign them to selected devices or groups. Once deployed, the devices automatically execute the configured tests and upload the results to the central server.

This approach improves consistency, reduces manual effort, and enables continuous network monitoring without requiring constant supervision.

7.1 Test Scheduling

The scheduling engine allows engineers to configure comprehensive test plans based on operational requirements.

A typical schedule may include:

- FTP Download Test
- FTP Upload Test
- HTTP Download Test
- HTTP Upload Test
- Voice Call Origination
- Voice Call Reception
- SMS Testing
- Ping Test
- Idle Mode Monitoring
- Data Session Monitoring

Schedules can specify:

- Start date
- Start time
- End time
- Test duration
- Test interval
- Number of repetitions
- Target server
- Network technology
- Assigned device group

Once configured, schedules are stored on the server and can be reused for future testing campaigns.

7.2 Real-Time Schedule Deployment

Centra-SD uses push technology to distribute schedules to field devices immediately after assignment.

This eliminates the need for manual configuration on individual smartphones.

When a schedule is updated, the latest configuration is automatically synchronized with all assigned devices.

Benefits include:

- Instant deployment
 - Consistent configuration
 - Reduced human error
 - Faster project execution
 - Centralized control
-

7.3 Continuous Monitoring

Centra-SD supports long-duration monitoring where measurements continue for hours, days, or even months.

Continuous monitoring is particularly useful for:

- Network stability evaluation
- Peak-hour analysis
- Weekend traffic analysis
- Holiday performance monitoring
- Service Level Agreement (SLA) verification
- Long-term optimization projects

Since measurements are fully automated, organizations can perform **24 × 7 × 365** monitoring with minimal operational overhead.

8. Remote Device Management

Managing hundreds of geographically distributed smartphones can become a significant operational challenge. Centra-SD simplifies this process by providing comprehensive remote device management capabilities through its web interface.

Administrators no longer need physical access to individual devices for routine operations.

8.1 Device Registration

Before participating in testing campaigns, each smartphone is registered within the Centra-SD platform.

During registration, information such as the following can be associated with the device:

- Device Name
- Device ID
- IMEI (if applicable)

- Assigned Operator
- Region
- Project
- Team
- Software Version
- Current Status

This information helps administrators organize and identify devices efficiently.

8.2 Device Grouping

Large organizations often manage hundreds of test devices. Centra-SD allows devices to be grouped according to operational requirements.

Example groups include:

- North Zone
- South Zone
- East Region
- West Region
- LTE Devices
- 5G Devices
- Indoor Testing Team
- Drive Test Vehicles
- Customer Complaint Investigation Team

Grouping simplifies schedule assignment and large-scale management.

8.3 Device Status Monitoring

Centra-SD continuously tracks the operational status of each connected device.

Typical status indicators include:

- Online
- Offline
- Active Test Running
- Idle
- Uploading Logs

- Synchronizing
- Battery Low
- Storage Full
- Network Disconnected

These indicators help administrators quickly identify devices requiring attention.

8.4 Remote Configuration

Administrators can remotely modify device configurations without requiring physical access.

Configuration options may include:

- Test profile selection
- Schedule updates
- Upload settings
- Logging intervals
- Server configuration
- Device grouping

This capability reduces maintenance time and allows rapid deployment of updated testing strategies.

9. KPI Monitoring

Key Performance Indicators (KPIs) provide quantitative measures of network performance. Centra-SD consolidates KPI data from all connected devices and presents it through intuitive dashboards.

This enables engineers to monitor network health at both macro and micro levels.

Common KPIs

Centra-SD can display numerous KPIs, including:

Radio Performance

- RSSI
 - RSRP
 - RSRQ
 - SINR
 - CQI
-

Mobility Performance

- Handover Success Rate
 - Handover Failure Rate
 - Cell Reselection
 - Attach Success Rate
-

Data Performance

- Download Throughput
 - Upload Throughput
 - Packet Loss
 - Latency
 - Jitter
-

Voice Performance

- Call Setup Success Rate
 - Call Drop Rate
 - Call Completion Rate
 - MOS Score
 - RTP Statistics
-

Device Performance

- Battery Level
 - Storage Utilization
 - GPS Status
 - Device Connectivity
-

10. Interactive Dashboard

Centra-SD provides an intuitive dashboard that serves as the central monitoring interface.

The dashboard summarizes the most important network statistics and enables engineers to monitor testing activities in real time.

Typical dashboard widgets include:

- Active Devices
- Devices Offline
- Running Tests
- Completed Tests
- Failed Tests
- Current Throughput
- Average Signal Strength
- Average Latency
- Voice Quality Indicators

These widgets provide an overview of the testing network without requiring detailed analysis.

11. Interactive Map Visualization

One of the most powerful features of Centra-SD is its map-based visualization.

Every uploaded measurement contains location information that can be displayed on an interactive map.

This enables engineers to understand how network performance varies geographically.

Coverage Visualization

Measurements can be displayed as color-coded points representing signal quality.

Typical colors include:

- Green – Excellent Coverage
- Yellow – Moderate Coverage
- Orange – Weak Coverage
- Red – Poor Coverage

This visualization allows engineers to identify coverage gaps quickly.

Issue Localization

The map allows users to zoom to approximately **100-meter resolution**, making it easier to identify the exact location where network problems occur.

This level of detail supports:

- Coverage optimization
- Cell planning

- Site verification
 - Customer complaint investigation
-

Route Playback

Drive-test routes can be replayed using uploaded GPS coordinates.

Engineers can observe:

- Device movement
- Signal variation
- Handover locations
- Throughput changes
- Voice quality along the route

This capability is particularly useful for mobility optimization.

12. Charts and Graphical Analysis

While maps provide geographical insight, charts help engineers analyze performance trends over time.

Centra-SD provides multiple chart types, including:

- Line Charts
- Bar Charts
- Pie Charts
- Histogram Views
- Trend Analysis Graphs

These visualizations simplify interpretation of large measurement datasets.

Time-Based Analysis

Engineers can evaluate KPI variations throughout the day.

Examples include:

- Morning traffic
- Peak-hour congestion
- Evening performance
- Weekend utilization

Time-based analysis assists operators in capacity planning and performance optimization.

Comparative Analysis

Charts can compare:

- Different operators
- Multiple devices
- Various technologies (5G vs LTE)
- Geographic regions
- Different testing periods

This enables benchmarking of network performance across multiple dimensions.

13. Data Filtering and Drill-Down Analysis

Large-scale testing campaigns generate substantial amounts of measurement data. Centra-SD provides advanced filtering capabilities that allow engineers to focus on specific subsets of information.

Common filter criteria include:

- Date Range
- Time Range
- Region
- Device
- Operator
- Network Technology
- KPI Threshold
- Signal Strength
- Test Type

Filters help reduce information overload and improve analysis efficiency.

Drill-Down Analysis

After identifying an issue through summary dashboards, users can drill down into individual measurements for detailed investigation.

Examples include:

- Viewing Layer 3 protocol messages

- Inspecting throughput variations
- Examining call failure events
- Reviewing GPS location
- Opening individual log files
- Comparing multiple KPIs simultaneously

This hierarchical analysis enables engineers to transition seamlessly from high-level monitoring to detailed troubleshooting.

14. Pre-Analysis Features

Centra-SD includes pre-analysis tools that automatically identify potential problem areas before engineers begin detailed investigations.

Examples include:

- Areas with consistently poor signal strength
- Frequent call drops
- High packet loss regions
- Low throughput locations
- Devices experiencing repeated failures

By highlighting anomalies in advance, pre-analysis reduces the time required to locate and resolve network issues.

15. Historical Performance Analysis

One of the most valuable capabilities of Centra-SD is its ability to maintain a centralized historical repository of all uploaded measurement data. Rather than analyzing network performance based solely on real-time information, engineers can compare current measurements with historical records to identify long-term trends, recurring issues, and the effectiveness of network optimization activities.

Historical analysis enables telecom operators to understand how network performance evolves over time. By examining measurements collected over weeks or months, engineers can determine whether network quality has improved, remained stable, or degraded after infrastructure changes or increased subscriber activity.

The centralized repository also supports compliance reporting, service level agreement (SLA) verification, and long-term performance benchmarking.

Benefits of Historical Analysis

- Compare current and previous network performance
- Monitor long-term KPI trends

- Evaluate optimization effectiveness
- Detect recurring network issues
- Support SLA compliance reporting
- Generate historical performance reports
- Identify seasonal or time-based traffic patterns

Historical data is especially valuable during network expansion projects, where engineers need to compare performance before and after deploying new base stations or upgrading existing infrastructure.

16. Data Export and Reporting

Centra-SD allows engineers to export collected measurement data for advanced analysis and reporting. Export functionality enables integration with third-party analytical tools and simplifies collaboration among different engineering teams.

Supported Export Formats

The platform supports exporting data in commonly used industry-standard formats, allowing compatibility with various reporting and visualization applications.

Common export formats include:

- CSV (Comma-Separated Values)
- Excel-compatible formats
- PDF Reports
- Measurement Log Files
- DLF (Drive Log Format)

These formats enable users to create custom reports, perform statistical analysis, or archive measurement data for future reference.

Sigma-PA Integration

Centra-SD integrates seamlessly with **Sigma-PA**, allowing engineers to perform more detailed post-processing of exported measurement logs.

Using the one-click export feature, selected measurement files can be transferred directly to Sigma-PA for advanced protocol analysis, troubleshooting, and reporting.

This integration reduces manual effort and ensures a smooth workflow between field measurements, centralized monitoring, and in-depth analysis.

Typical workflow:

1. Field device performs automated tests.

2. Data is uploaded to Centra-SD.
 3. Engineers identify areas requiring deeper investigation.
 4. Relevant logs are exported.
 5. Sigma-PA performs advanced analysis.
 6. Reports are generated for optimization teams.
-

17. Central Repository Architecture

Centra-SD functions as a centralized repository for all measurement data collected from remote test devices. Every uploaded log is securely stored and organized based on configurable attributes such as project, region, operator, device, and date.

A centralized repository provides several advantages:

- Eliminates scattered log files across multiple devices
- Simplifies data retrieval
- Supports multi-user collaboration
- Enables centralized backup
- Improves data consistency
- Facilitates historical analysis

Because all measurements are stored in a single location, engineering teams can access the same dataset, ensuring consistent analysis and decision-making across the organization.

18. Integration with Sigma-ML

Centra-SD is designed to work closely with Sigma-ML, forming a complete wireless network testing ecosystem.

Sigma-ML

Sigma-ML is responsible for performing field measurements on smartphones. It collects radio parameters, protocol messages, throughput statistics, voice quality metrics, and GPS information.

Centra-SD

Centra-SD receives measurement data uploaded by Sigma-ML devices, stores it in a centralized repository, visualizes KPIs, manages devices, and coordinates testing operations.

Combined Workflow

1. Network administrator creates test schedules in Centra-SD.
2. Schedules are pushed to Sigma-ML devices.
3. Sigma-ML executes the assigned measurements automatically.

4. Measurement logs are uploaded to Centra-SD.
5. Centra-SD consolidates and visualizes the data.
6. Engineers analyze KPIs, identify issues, and export logs if required.
7. Optimization teams implement corrective actions.
8. The testing process is repeated to verify improvements.

This integrated workflow enables efficient network monitoring and optimization with minimal manual intervention.

19. Security and User Access

As a centralized platform, Centra-SD incorporates mechanisms to control user access and protect measurement data.

User Authentication

Only authorized users can access the platform. Authentication ensures that sensitive network information is available only to approved personnel.

Role-Based Access

Different user roles may be assigned based on operational responsibilities.

Examples include:

- Administrator
- Network Engineer
- Field Technician
- Project Manager
- Read-Only Analyst

Each role can be configured with appropriate permissions for viewing, modifying, or exporting data.

Secure Data Storage

Measurement logs are stored in the central repository, where they can be backed up regularly to prevent data loss and ensure business continuity.

20. Practical Use Cases

Centra-SD supports a wide variety of operational scenarios within telecommunications organizations.

Nationwide Drive Testing

Operators managing nationwide cellular networks can monitor hundreds of drive-test devices simultaneously through a single dashboard.

Network Optimization

Optimization teams can analyze uploaded KPI data to identify weak coverage areas, failed handovers, or throughput issues and implement corrective actions.

Customer Complaint Investigation

When subscribers report poor service quality, engineers can locate measurements collected near the reported area and analyze historical performance to determine the underlying cause.

New Network Rollout

During deployment of new LTE or 5G sites, Centra-SD provides centralized monitoring of acceptance tests, ensuring that performance objectives are met before commercial launch.

Vendor Benchmarking

Telecom operators can compare the performance of different equipment vendors, technologies, or geographic regions using standardized KPI dashboards and reports.

Research and Development

Research institutions and universities can use Centra-SD to collect, manage, and analyze wireless network data for experimental studies and technology evaluations.

21. Advantages of Centra-SD

Centra-SD offers several operational and business advantages that improve the efficiency of wireless network monitoring.

Centralized Management

All devices, schedules, and measurement data are managed from a single location, reducing administrative complexity.

Increased Operational Efficiency

Automation reduces manual effort, enabling engineers to focus on analysis rather than repetitive operational tasks.

Scalability

The platform supports deployment across large organizations with numerous remote devices, making it suitable for nationwide or enterprise-scale monitoring.

Real-Time Visibility

Live dashboards provide immediate insight into network conditions, allowing engineers to respond quickly to performance degradation.

Historical Analysis

Centralized storage enables comparison of current and historical measurements, supporting long-term optimization strategies.

Improved Collaboration

Multiple users can access the same repository, promoting collaboration between field engineers, analysts, and management teams.

Reduced Operational Costs

Browser-based access and smartphone-based field testing reduce the need for expensive hardware and simplify deployment.

22. Best Practices

To maximize the effectiveness of Centra-SD, organizations should follow several recommended practices.

- Organize devices into logical groups based on project or geography.
 - Create reusable test schedules for recurring measurement campaigns.
 - Monitor battery and storage status of field devices regularly.
 - Verify successful data uploads after each measurement session.
 - Review KPI dashboards periodically to identify emerging issues.
 - Archive historical logs for future reference and compliance reporting.
 - Use filters and drill-down analysis to investigate abnormal KPI values.
 - Export important measurement logs for detailed analysis in Sigma-PA.
 - Regularly back up the centralized repository.
 - Keep both the server and field devices updated with the latest software versions.
-

23. Frequently Asked Questions (FAQs)

What is Centra-SD?

Centra-SD is a centralized web-based platform for managing remote wireless network test devices, monitoring network performance, scheduling automated tests, and analyzing collected measurement data.

Does Centra-SD perform field measurements directly?

No. Field measurements are performed by applications such as Sigma-ML running on smartphones. Centra-SD manages these devices and analyzes the uploaded data.

Can Centra-SD manage multiple devices simultaneously?

Yes. The platform is designed to manage an unlimited number of registered test devices, depending on server capacity and licensing.

Is internet connectivity required?

Yes. Remote management, schedule deployment, and data upload require internet connectivity between field devices and the Centra-SD server.

Can historical data be analyzed?

Yes. All uploaded measurements are stored in a centralized repository, enabling historical performance analysis and long-term trend evaluation.

Which browsers are supported?

Centra-SD is accessible through standard web browsers, allowing users to monitor the platform without installing additional client software.

24. Glossary

KPI (Key Performance Indicator): A measurable value used to evaluate network performance.

FTP (File Transfer Protocol): A protocol used for testing file upload and download performance.

HTTP (Hypertext Transfer Protocol): A protocol commonly used to evaluate web application performance.

Latency: The time taken for data to travel between two points in a network.

Throughput: The amount of data successfully transmitted over a network within a given period.

Drive Testing: The process of collecting wireless network measurements while moving through a geographic area.

Central Repository: A centralized storage location where all measurement data is collected and maintained.

Push Technology: A communication mechanism used to distribute schedules and configuration updates from the server to field devices in real time.

25. Conclusion

Centra-SD is a comprehensive centralized management and analytics platform that simplifies large-scale wireless network monitoring. By combining automated scheduling, remote device management, real-time KPI visualization, historical data analysis, and seamless integration with Sigma-ML, the platform enables organizations to efficiently monitor, evaluate, and optimize mobile network performance.

Its browser-based interface, centralized architecture, and powerful analytical capabilities reduce operational complexity while providing engineers with the tools needed to identify network issues quickly and accurately. Whether deployed for nationwide drive testing, indoor coverage verification, network optimization, or long-term performance monitoring, Centra-SD offers a scalable and efficient solution for managing distributed wireless testing operations.

For organizations seeking to improve network quality, streamline field testing, and enhance operational efficiency, Centra-SD serves as a robust foundation for centralized wireless network performance management.